Natural Language Processing
Project Documentation

1. Students Information

Team Members	Shahd Ramadan - Baina Magdy – Nourhan Mohamed – Israa Alaa
GitHub	https://github.com/bainamagdy/NLP-Project
Deployment Link	https://bainamagdy-nlp-project-app-pw8j7h.streamlit.app/
Department / Major	Faculty of Computers & Artificial Intelligence / Computer Science
Academic Year	2025 / 2026

2. Project Overview

Project Title	Spam Email Classifier (Hybrid BiLSTM-Attention & ML Approach)
Submission Date	May 29, 2026

Project Summary

This project develops a spam email classification system that combines traditional machine learning models with deep learning techniques. Three models are implemented and compared: Linear SVM, XGBoost, and a Bidirectional LSTM with an Attention mechanism. The system processes both text content (via TF-IDF vectorization with bigrams) and 12 structured metadata features (e.g., sender domain, number of links, urgency terms). A Streamlit web application is built to allow users to test any email in real time. All three models achieve near-perfect accuracy on the test set.

3. NLP Topic Area

Select the NLP topic(s) your project will focus on:

☑ Text Classification ☐ Sentiment Analysis ☐ Named Entity Recognition

☐ Machine Translation ☐ Text Generation / LLMs ☐ Speech Processing

☐ Question Answering ☐ Text Summarization ☑ Other (specify below)

If 'Other', specify the NLP topic:

The project uses feature engineering + metadata fusion (sender domain, link counts, urgency terms, timestamps) combined with deep learning (BiLSTM + Attention), which goes beyond standard NLP text classification into a hybrid ML/NLP pipeline not captured by the other options.

4. Problem Statement

Problem Statement

Email spam remains a significant cybersecurity and productivity challenge. Millions of spams and phishing emails are sent daily, causing financial losses, data breaches, and wasted time. Existing rule-based filters often fail against sophisticated, context-aware spam that mimics legitimate communication. There is a need for an intelligent, adaptive system that can accurately identify spam by analyzing both the textual content and behavioral metadata of emails.

Motivation & Relevance

Spam detection is a highly relevant NLP application with direct real-world impact. This project demonstrates the practical integration of classical NLP (TF-IDF, lemmatization, stopword removal) with modern deep learning (BiLSTM + Attention). It highlights the value of combining text-based and metadata features for improved classification, and addresses real challenges such as feature leakage, label encoding of categorical variables, and calibration of classifier probabilities.

5. Proposed Methodology

Programming Language	Python
Frameworks / Libraries	Streamlit, Pandas, NumPy, Scikit-Learn, XGBoost, TensorFlow, Keras, NLTK, SciPy, Joblib, Seaborn, Matplotlib
Dataset / Link	https://www.kaggle.com/datasets/algozee/spam-vs-legitimate-email-dataset/data

Data Collection & Preprocessing

The dataset is loaded from a CSV file containing email text, subject, and metadata. Preprocessing steps include: (1) dropping redundant/non-predictive columns (email_id, has_attachment, num_characters, num_exclamation_marks); (2) combining subject and email body into a single full_text field; (3) lowercasing, punctuation removal, and digit removal; (4) POS-aware lemmatization using WordNet; (5) stopword removal; (6) encoding the sender_domain column using LabelEncoder; (7) TF-IDF vectorization with bigrams (max 3,000 features, min_df=2); (8) StandardScaler normalization of numeric features; (9) a feature leakage audit before training to detect inflated accuracy from metadata-only predictors.

Model / Algorithm Approach

Three models are trained on a combined feature matrix (TF-IDF text + 12 scaled numeric features): (1) Linear SVM wrapped with CalibratedClassifierCV for probability output; (2) XGBoost with hist tree method for sparse matrix efficiency; (3) a hybrid BiLSTM + custom Attention Layer model that takes padded token sequences (max length 150, vocab 5,000) and scaled numeric features as dual inputs. The deep learning model uses early stopping (patience=3) and is saved in the modern .keras format.

Evaluation Metrics

Accuracy, Precision, Recall, F1-Score (via classification_report). Cross-validation F1 (5-fold) for SVM and XGBoost. Results: Linear SVM — 100% accuracy; XGBoost — 99.95% accuracy; BiLSTM + Attention — 100% accuracy.

6. Project Framework

7. Expected Challenges & Mitigation

Anticipated Challenges

1. Feature leakage: metadata columns such as sender_reputation_score may directly encode the label, causing artificially high accuracy. 2. Out-of-vocabulary domains: the LabelEncoder may encounter unseen sender domains at inference time. 3. Class imbalance: if spam/ham ratios are skewed, models may be biased. 4. Deep learning convergence: BiLSTM training stability and overfitting on small datasets. 5. Deployment dependencies: ensuring all saved model artifacts (.keras, .joblib) are correctly loaded in the Streamlit app.

Mitigation Plan

1. Leakage audit: a GradientBoostingClassifier trained solely on metadata is run before full training; accuracy above 90% signals leakage. 2. Safe encoding: unknown domains are mapped to 0 at inference. 3. Stratified split: train_test_split uses stratify=y to preserve class balance. 4. Early stopping: monitors validation loss with patience=3 and restores best weights. 5. Artifact versioning: all models, vectorizers, scalers, and tokenizers are saved via joblib and the Keras .keras format for reliable loading.

8. Preliminary References

List at least 15 references (papers, datasets, tools) you plan to use. Use APA format.

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